PROJECT STAGE - II

FORENSIC FACE SKETCH RECOGNITION AND CONSTRUCTION

A dissertation submitted in partial fulfillment of the

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BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

Submitted by

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This is to certify that the Project Report entitled "**Forensic Face Sketch Recognition and Construction"** is a bonafide work done and submitted by **T. Chidwila (22B81A1271)**, **P. Maithri (22B81A1287), T. Sri Varsha (22B81A12B7)** during the academic year 2025-2026, in partial fulfillment of requirement for the award of Bachelor of Technology degree in Information Technology from Jawaharlal Nehru Technological University Hyderabad, is a bonafide record of work carried out by them under my guidance and supervision.

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DECLARATION

We hereby declare that the project report entitled **Forensic Face Sketch Recognition and Construction** is an original work done and submitted to the IT Department, CVR College of Engineering, affiliated to Jawaharlal Nehru Technological University Hyderabad, Hyderabad in partial fulfillment of the requirement for the award of Bachelor of Technology in **Information Technology** and it is a record of bonafide project work carried out by us under the guidance of **Mrs. G. Shailaja**, Assistant Professor, **Department of Information Technology.**

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ABSTRACT

In criminal investigations, hand-drawn face sketches based on eyewitness descriptions have traditionally played a vital role in identifying suspects. However, in today's fast-paced digital age, manually drawn sketches present significant limitations when it comes to efficient and accurate suspect recognition from large-scale databases. Traditional sketch-based identification methods and earlier software applications have shown limited effectiveness due to constraints such as predefined facial feature sets, low realism, and poor integration with automated recognition system. This project proposes an advanced face sketch construction and recognition platform powered by deep learning and cloud infrastructure. Unlike previous approaches, our application allows users not only to select facial features from an expanded set but also to upload hand-drawn individual features. These are then transformed into application compatible components, ensuring greater accuracy and resemblance to actual sketches. Additionally, the platform enables law enforcement agencies to upload entire sketches, which can be matched against real-time or existing databases using intelligent matching algorithms. By leveraging machine learning, the system suggests compatible features and optimizes the recognition process - enhancing both speed and accuracy in suspect identification.

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CHAPTER – 1

INTRODUCTION

1.1 Motivation

In recent years, the rate of criminal activities has been rising, and law enforcement agencies are under pressure to solve cases faster and more accurately. Traditionally, identifying a suspect often relied on hand-drawn sketches made by forensic artists based on witness descriptions. While this method has helped in many cases, it also has several challenges — it's time-consuming, depends heavily on the artist's skill, and isn't always accurate when compared to large databases of faces.

This situation motivated us to create a **Forensic Face Sketch Construction and Recognition System** — a smart and modern platform that combines traditional sketching methods with advanced technology. Our goal is to make the process of creating and recognizing suspect faces faster, more reliable, and easier for law enforcement officers.

The main motivation behind this project is to:

- Help officers identify suspects quickly and with higher accuracy.
- Reduce the dependency on manual sketching and artist availability.
- Provide an easy-to-use interface where users can drag and drop facial features to build sketches.
- Improve data security and privacy through features like two-step verification and machine locking.
- Use deep learning and cloud computing to match sketches with large face databases efficiently.

In short, we were driven by the idea of bringing technology and innovation into the investigation process — making it easier for the police to find suspects, save time, and improve the overall effectiveness of criminal identification.

1.2 Problem Statement

As technology continues to advance, the methods used by law enforcement agencies to identify and capture criminals must also evolve. Despite the availability of modern tools, one of the oldest and most widely used techniques in criminal investigations — the creation of suspect sketches based on witness descriptions — still faces many challenges. Traditionally, these sketches are hand-drawn by forensic artists and then circulated to help identify suspects. While this approach has been effective in the past, it is now proving to be slow, inconsistent, and less compatible with modern digital systems.

The major challenge lies in the **manual and time-consuming nature** of this process. Forensic sketch artists require time and skill to translate a witness's verbal description into an accurate visual sketch. Moreover, there are very few trained forensic artists available compared to the rising number of cases, leading to delays in investigations. Another major limitation is the **difficulty of matching hand-drawn sketches** with digital facial images stored in police databases. Since traditional sketches and digital photos differ in format, texture, and detail, automatic recognition systems often fail to achieve the desired accuracy.

Previous attempts to automate this process have not been entirely successful. Some applications allowed users to assemble faces by choosing from a limited set of facial features, but the results often looked unrealistic or cartoon-like, making them unsuitable for professional use. Others focused only on recognition but lacked flexibility for sketch creation. These shortcomings have created a **gap between traditional sketching methods and modern recognition technologies.**

Therefore, there is a clear need for a **comprehensive digital platform** that can both construct and recognize forensic face sketches efficiently. Such a system should allow law enforcement officers to either upload existing hand-drawn sketches or create new ones digitally using predefined facial components. The platform should also integrate **machine learning and deep learning algorithms** to enhance recognition accuracy by comparing sketches with large criminal databases in real time.

By addressing these issues, the proposed system aims to **modernize the forensic sketching process**, reduce investigation time, and increase the accuracy of suspect identification, ultimately helping law enforcement agencies deliver faster and more reliable justice.

1.3 Project Objectives

The main goal of this project is to create a smart and user-friendly **Forensic Face Sketch Construction and Recognition System** that can help law enforcement officers identify criminals quickly and accurately. The idea is to combine traditional forensic sketching methods with modern technologies like artificial intelligence, deep learning, and cloud computing to make the investigation process faster, simpler, and more reliable.

Our system aims to allow users to either **create a suspect's face digitally** using predefined facial features such as eyes, nose, lips, and ears, or **upload an existing hand-drawn sketch** to find a match in the criminal database. This removes the need for expert sketch artists and makes it possible for anyone in the department to use the tool easily.

The main objectives of our project are:

- To replace the slow, manual sketching process with a **modern digital platform**.
- To use **AI and deep learning** to automatically compare sketches with stored images of known criminals.
- To maintain **high levels of security and privacy** through features like two-step login verification and system-based access restrictions.
- To make the system **backward compatible**, so even old hand-drawn sketches can be uploaded and recognized.
- To offer an **easy drag-and-drop interface** that helps officers construct accurate sketches quickly, even without artistic skills.
- To **save time and effort** by speeding up the suspect identification process.
- To use **cloud services (like AWS)** for storing and processing data safely and efficiently.

In short, the objective of this project is to **help law enforcement agencies modernize their investigation methods** by using advanced technology to identify suspects more efficiently, ensure data security, and improve the overall accuracy of criminal identification.

1.4 Project Report Organization

Abstract:

This section provides a summary of the *Forensic Face Sketch Construction and Recognition System* project, including its motivation, problem statement, objectives, and outcomes. The project aims to assist law enforcement agencies in identifying criminals quickly and accurately using a hybrid system that combines traditional forensic sketching with modern technologies like deep learning and cloud computing. The application enables users to construct digital facial sketches through a drag-and-drop interface and recognize suspects by matching sketches with criminal databases. Key outcomes include improved identification accuracy, enhanced data security through two-step authentication and machine locking, and seamless compatibility with both hand-drawn and digital sketches.

Introduction:

The introduction highlights the increasing challenges faced by law enforcement in criminal identification due to the time-consuming nature of manual forensic sketching and the shortage of skilled artists. It discusses how this project bridges the gap between traditional methods and modern technology by using a deep learning–based recognition system. The section also outlines the motivation, problem statement, and objectives behind developing a secure, user-friendly, and cloud-integrated platform for face sketch creation and recognition.

Literature Survey:

This section reviews previous research and techniques in the field of forensic face sketch construction and recognition. It summarizes various methods proposed by researchers such as Charlie Frowd, Tang & Wang, and others, focusing on their limitations—like low accuracy, poor realism, and difficulty in matching sketches to photos. The review emphasizes the need for a system that can handle both hand-drawn and digitally constructed sketches while offering higher precision and speed through the use of deep learning and centralized computing.

Software Reuirement Specifications:

This section explains the key design considerations of the system, focusing on security, privacy, and backward compatibility. It introduces features such as machine locking (based on MAC and hard drive IDs), two-step verification for secure login, and centralized server connectivity for data protection. It also discusses how the system supports both new digital sketches and old hand-drawn ones, ensuring smooth adoption within law enforcement agencies.

Design:

This part details the overall design and workflow of the application. It covers how users log in securely, create face sketches using a drag-and-drop interface, and then match those sketches with stored records in the database. It describes two main modules — the Face Sketch Construction Module and the Face Sketch Recognition Module — supported by cloud infrastructure and machine learning algorithms. Flowcharts, system diagrams, and user interface layouts are included to illustrate the design and functionality.

Implementation:

This section demonstrates the working of the application through step-by-step screenshots, from the splash screen and login verification to the dashboard for sketch creation and recognition. It explains how users can construct facial sketches by selecting facial features and then use the recognition module to find matching suspects. Visuals such as flowcharts, face feature panels, and database structures support the explanation of the complete implementation process.

Testing:

The testing phase of the Forensic Face Sketch Construction and Recognition System was successfully completed by evaluating all modules with various input scenarios. The login module ensured accurate authentication by validating different username and password combinations. The face sketch construction module worked efficiently, allowing users to create and save sketches without errors. The face recognition module accurately processed sketches, extracted features, and matched them with database records, providing reliable similarity results. The system also handled invalid inputs and unmatched cases effectively without crashing. Overall, the system proved to be stable, secure, and suitable for real-world law enforcement applications.

Conclusion:

The project achieved strong results in both performance and security. Testing showed over 90% accuracy in face recognition and 100% reliability . The system effectively identified suspects from both hand-drawn and digitally constructed sketches, demonstrating its practicality for real-world use. The conclusion highlights how the platform enhances investigation efficiency, ensures data security, and bridges the gap between traditional forensic sketching and modern AI-driven methods.

Future Enhancement:

This section discusses possible future enhancements, including integrating 3D facial mapping, live CCTV feed analysis, and social media data to expand the system's capabilities. It also suggests upgrading the recognition module to work in real-time environments and connecting the platform with global criminal databases for broader reach and faster results.

References:

Includes cited research papers, books, and articles on face sketch recognition, deep learning algorithms, forensic identification systems, and cloud-based applications used in law enforcement.

CHAPTER – 2
LITERATURE SURVEY

2.1 EXISTING WORK

Research on **forensic face sketch construction and recognition** has evolved significantly over the past few decades, aiming to improve how law enforcement identifies suspects based on eyewitness descriptions. Earlier, sketches were manually drawn by forensic artists and compared visually against criminal databases, but this process was slow, subjective, and often inaccurate. Over time, researchers introduced digital and algorithmic solutions to automate sketch construction and recognition. These methods can be broadly classified into three major areas:

- **Traditional face composite systems** based on manual or semi-automated assembly of facial features,
- **Algorithm-based photo-sketch synthesis and recognition techniques**, and
- **Machine learning and deep learning-based approaches** for improving precision and automation.

Representative works and their findings are discussed below.

2.2 LIMITATIONS OF EXISTING WORK

While previous studies and applications in forensic face sketch construction and recognition have made important progress, several limitations still restrict their real-world effectiveness. Most existing systems either focus only on creating sketches or on recognizing them, but very few combine both in a secure, practical, and efficient way suitable for law enforcement. The key limitations identified from earlier research are summarized below:

2.2.1. Limited Accuracy and Realism

1. **Traditional composite tools** like FACES and Identi-Kit rely on predefined facial parts, which often result in unrealistic, cartoon-like sketches.
2. The generated sketches lack natural facial blending and lifelike proportions, making them hard to match with real photographs in police databases.
3. Manual sketch-based recognition techniques depend heavily on the artist’s skill and the witness’s description, leading to inconsistent and subjective results.

2.2.2. Ineffective Matching and Recognition Algorithms

1. Many earlier methods such as **SIFT descriptors**, **Markov Random Field (MRF)** models, or **feature-based comparisons** work only under ideal conditions, like front-facing images with consistent lighting.
2. These algorithms often fail when sketches and photos differ in pose, angle, or facial expression — a common real-world scenario.
3. Accuracy drops significantly when dealing with low-quality sketches or limited datasets, limiting their use for large-scale police operations.

2.2.3. Lack of Integration Between Traditional and Modern Techniques

1. Most existing systems are **not backward compatible**, meaning they cannot process or recognize traditional hand-drawn sketches.
2. There is no unified platform that allows both **sketch construction and recognition** within a single application.
3. Law enforcement agencies face challenges transitioning from manual methods to digital systems due to the absence of such hybrid solutions.

2.2.4. Poor User Accessibility and System Complexity

1. Many earlier applications are designed for technical or forensic experts, requiring specialized training.
2. Complex interfaces and rigid workflows make them difficult for regular officers or non-technical users to operate.
3. The lack of drag-and-drop functionality or easy customization slows down the suspect sketching process during critical investigations.

2.2.5. Security and Privacy Concerns

1. Previous research often ignored **system-level security**, leaving sensitive criminal data vulnerable to misuse or unauthorized access.

2. Most older systems lack **authentication mechanisms** like hardware-based machine locking.
3. Without centralized control or encryption, these tools are unsuitable for deployment in official law enforcement networks.

2.2.6. Limited Computational Efficiency and Scalability

1. Traditional methods are **time-consuming** and not optimized for large databases or real-time face recognition.
2. Lack of cloud-based infrastructure limits their performance and scalability.
3. Many systems are standalone desktop tools that cannot process or compare data from distributed sources.

2.2.7. Absence of AI-Based Learning and Automation

1. Earlier systems do not use **machine learning or deep learning algorithms** to improve with experience or suggest related facial features automatically.
2. This makes the sketching process slower and less intelligent, with minimal automation or adaptability.
3. The absence of predictive algorithms prevents the system from recommending suitable facial features based on witness inputs.

2.2.8. Summary of Identified Gaps

In summary, existing forensic sketch systems suffer from:

- Low accuracy and poor realism in generated sketches.
- Weak matching performance for non-frontal or hand-drawn sketches.
- Lack of secure, cloud-enabled infrastructure for real-time use.
- Complex user interfaces and limited accessibility for law enforcement staff.
- Minimal use of AI, resulting in poor adaptability and automation.

These limitations highlight the need for a **modern, intelligent, and secure forensic platform** that combines sketch creation, recognition, and cloud-based storage — all within a single, user-friendly application.

2.3 Traditional Composite Face Construction Systems

Early research in forensic identification relied on manual sketching or basic computer-assisted systems. For instance, **Dr. Charlie Frowd and colleagues** developed an application that helped witnesses select facial features from a database to form a composite face. Although it reduced some manual effort, it still depended heavily on human judgment and produced sketches that often lacked realism and accuracy.

Similarly, early commercial systems like **Identi-Kit** and **FACES 4.0** provided limited facial feature libraries for constructing suspect faces. However, these systems had several shortcomings — the generated sketches appeared cartoon-like, lacked lifelike detail, and often failed to match real faces effectively. Witnesses also found it difficult to describe facial features precisely using pre-defined options, resulting in lower identification accuracy.

These limitations motivated the shift toward algorithm-driven systems that could better synthesize and compare facial sketches with real images.

2.3.1 Algorithmic and Recognition-Based Methods

Later studies focused on transforming hand-drawn sketches into photo-like images for easier comparison with mugshots or surveillance photos. **Xiaoou Tang and Xiaogang Wang** proposed a *Multiscale Markov Random Field (MRF)* model to synthesize sketches into photos and vice versa. Their approach divided a face into small patches and reduced the difference between sketch and photo domains. While this improved matching accuracy, it struggled when sketches and photos were captured at different angles.

Anil K. Jain and Brendan Klare introduced a *feature-based matching* approach using the **SIFT Descriptor**, measuring the similarity between sketches and photos based on local texture patterns. This method performed better than earlier pixel-based comparisons but still had difficulty handling facial orientation, lighting, and artistic variations.

P.C. Yuen and C.H. Man later proposed converting sketches into mugshots and then matching them to real faces using global and local variables. Although it achieved around **70% accuracy**, performance dropped significantly for non-frontal faces and low-quality sketches.

Overall, these algorithmic methods provided foundational progress in the field but were still limited in handling complex real-world conditions such as varied lighting, pose differences, and artistic inconsistencies.

2.3.2 Machine Learning and Deep Learning Approaches

The most recent developments in forensic sketch recognition involve **machine learning** and **deep learning**. These methods learn from large datasets of faces and sketches to automatically extract important features. Deep neural networks can now analyze both hand-drawn sketches and digital composites, recognizing patterns with high precision.

Deep learning models like **Convolutional Neural Networks (CNNs)** have shown promising results by automatically learning to distinguish facial features under different poses, lighting, and drawing styles. The introduction of large datasets such as *Labeled Faces in the Wild (LFW)* and *MegaFace Challenge* has further helped improve the training and evaluation of these models.

Despite their success, deep learning models require large computational power and training data. Real-world sketches may not always meet these requirements, leading to mismatches or limited performance in dynamic environments.

2.3.3 Methodological and Evaluation Trends

1. **Performance Metrics:**
   Most studies evaluate sketch recognition systems based on **accuracy**, **similarity score**, and **match percentage** with existing face databases. Accuracy tends to drop when faces are viewed from different angles or drawn by different artists.
2. **Feature Extraction and Comparison:**
   Techniques such as **SIFT descriptors**, **Local Binary Patterns (LBP)**, and **Haar features** have been used to improve recognition between sketches and real photos. Deep learning models now replace manual feature extraction by automatically identifying these facial landmarks.

3. **Database Diversity:**
   Research often uses standard datasets (FERET, CUFSF, CUHK) for evaluation. However, many studies lack variety in facial features and demographic representation, limiting real-world adaptability.
4. **Security and Deployment:**
   Few existing studies focus on **data privacy and secure access**. Most forensic applications overlook system-level security like login verification, encryption, and controlled access to criminal databases — all of which are critical for real deployment.

2.3.4 Short Critical Synthesis (What We Learn from Existing Work)

- **Manual and semi-automated systems** are simple but limited by human error, restricted feature sets, and unrealistic outputs.
- **Algorithmic and MRF-based models** improved realism but struggled with pose variations and lighting inconsistencies.
- **Deep learning-based systems** show the best performance so far, offering high accuracy and the ability to learn facial variations automatically.
- However, most research still lacks **secure integration, user-friendliness**, and **support for traditional hand-drawn sketches**. These gaps highlight the need for a unified, intelligent, and secure system for real-world forensic use.

2.3.5 Motivation

1. Existing systems are often **limited to research prototypes** and are not suitable for practical law enforcement environments.
2. **Most models cannot handle both digital and hand-drawn sketches**, which limits backward compatibility.
3. **Security and privacy mechanisms** such as authentication, machine locking, and centralized control are largely absent.
4. Many systems lack **user-friendly interfaces**, requiring technical or artistic expertise to operate.
5. There is still no **end-to-end platform** that combines sketch creation, feature selection, cloud-based recognition, and result visualization in a single secure environment.

2.3.6. Scope

The *Forensic Face Sketch Construction and Recognition System* directly addresses these limitations by developing a **complete, secure, and practical solution** for modern law enforcement.

This project introduces a platform that:

- Allows users to **construct facial sketches digitally** through drag-and-drop features.
- Supports **uploading traditional hand-drawn sketches** for backward compatibility.
- Uses **deep learning and AWS cloud infrastructure** for fast and accurate face matching.
- Ensures **high-level security and privacy** through machine locking.
- Provides an **intuitive interface** that requires no professional sketching skills.

By integrating both traditional and modern methods into one seamless platform, this project bridges the gap between **manual forensic techniques** and **AI-powered face recognition systems**, offering law enforcement a reliable and future-ready tool for criminal identification.

CHAPTER - 3

SOFTWARE REQUIREMENT SPECIFICATIONS

This SRS defines the functional and non-functional requirements for the Forensic Face Sketch Recognition and Construction System — a desktop and server-based application designed to assist law enforcement agencies in efficiently constructing composite face sketches from eyewitness descriptions and recognizing suspects by matching sketches against criminal databases using advanced deep learning techniques. The objective is to provide an unambiguous, testable, and implementation-ready description that ensures security, accuracy, and ease of use for developers, testers, forensic experts, and evaluators.

3.1 Functional Requirements

Each requirement is testable and uniquely identified.

1. **User Authentication and Security**
    - Secure login with two-step verification using registered email.
    - Machine locking using hardware and software parameters (hard drive ID, MAC address, IP address) to prevent unauthorized use.
    - Centralized user management linked to law enforcement department databases.
2. **Face Sketch Construction**
    - Provide a canvas with predefined facial feature sets (head, eyes, nose, lips, ears, etc.) for constructing composite sketches.
    - Enable drag-and-drop feature for resizing and repositioning facial elements according to eyewitness description.
    - Allow uploading of hand-drawn sketches or individual facial features to convert into application components.
    - Save constructed sketches in standard image formats (e.g., PNG, JPG)
3. **Face Sketch Recognition**
    - Upload constructed or hand-drawn face sketches to a centralized law enforcement server.
    - Map features of the sketch and match with the criminal database using deep learning algorithms.

- Display matched faces with similarity percentage, confidence values, and additional user metadata.
- Export and share match result metadata securely

4. **Machine Learning Enhancements**

- Use deep learning techniques to suggest relatable facial features while constructing a sketch to speed up the process.
- Use cloud infrastructure (AWS) for processing recognition tasks to enhance speed and accuracy.

5. **Database Management**

- Manage criminal records including face photos in a secure, centralized database with multi-layer access control.
- Maintain user credentials and hardware binding information securely.

6. **Compatibility**

- Backward compatibility with existing hand-drawn sketches allowing law enforcement to transition smoothly without losing previous data usage.

7. **Sketch Enhancement and Preprocessing**

- The system shall provide automated image preprocessing capabilities including edge enhancement, noise reduction, and adaptive filtering to improve sketch quality before recognition processing.
- The system shall automatically standardize uploaded sketches through resizing, normalization, and colour intensity adjustments to ensure uniformity for CNN processing.

8. **Feature Library Management**

- The system shall allow authorized users to add new facial features and components to the predefined feature library, expanding the available options for sketch construction.
- The system shall categorize and organize facial features by type (head shapes, eye types, nose variations, etc.) with search and filter capabilities for quick access during sketch creation.

9. **Multi-Format Export and Import**
    - The system shall support importing hand-drawn sketches in multiple image formats (JPEG, PNG, BMP, TIFF) and convert them for processing.
    - The system shall export completed sketches and recognition results in various formats including high-resolution images, PDF reports, and structured data files for integration with other law enforcement systems.
10. **Real-Time Processing and Batch Operations**
    - The system shall provide real-time sketch recognition capabilities with live feedback during the matching process, displaying confidence levels and potential matches as they are identified.
    - The system shall support batch processing functionality to analyse multiple sketches simultaneously against the database, enabling efficient processing of multiple cases.

3.2. Non-Functional Requirements

1. **Security**
    - The application must ensure data privacy and secure access by locking the application to specific hardware.
    - All uploaded data and recognition processes run on secure, centralized servers with controlled access.
2. **Usability**
    - User-friendly interface designed for non-professional users.
    - Fast processing by leveraging cloud resources.
    - System should respond promptly to user inputs and upload/download actions.
3. **Reliability and Availability**
    - Application should operate reliably on specified minimum hardware requirements (Intel dual-core, 1GB RAM client; Intel i3, 4GB RAM server).
    - Centralized server and cloud must maintain high availability to prevent downtime.

- Display matched faces with similarity percentage, confidence values, and additional user metadata.
- Fail-safe mechanisms to lock system if hardware tampering is detected.

4. **Scalability**
    - Ability to handle growing datasets of criminal records and sketches.
    - Cloud-based architecture to allow scaling computation resources as required.
5. **Compatibility**
    - Runs on Windows 7 and above.
    - Java-based desktop application using JavaFX for cross-platform GUI consistency.
6. **Performance**
    - Efficient deep learning algorithms to provide accurate results with average accuracy over 90% and high confidence levels.
    - Fast processing by leveraging cloud resources.
    - System should respond promptly to user inputs and upload/download actions.
7. **Maintainability**
    - Modular design for separate face construction and recognition components.
    - Use of standard libraries and frameworks (Java JDK, AWS CLI) for easier updates and maintenance.

3.3 Software and hardware Requirements

- **Frontend:** JavaFX, FXML Scene Builder, CSS styling, Canvas API
- **Backend:** Java JDK, Deep Learning frameworks (CNN models), SIFT Descriptors, Feature extraction algorithms, AWS SDK
- **Database:** SQLite, AWS Cloud Storage, Centralized law enforcement database system.
- **Cloud Services:** Amazon Web Services (AWS), AWS CLI, EC2 instances, S3 storage.

- **Machine Learning:** Convolutional Neural Networks (CNN), Deep Learning models for face recognition, Feature mapping algorithms.

CHAPTER - 4

DESIGN

4.1 Software Architecture

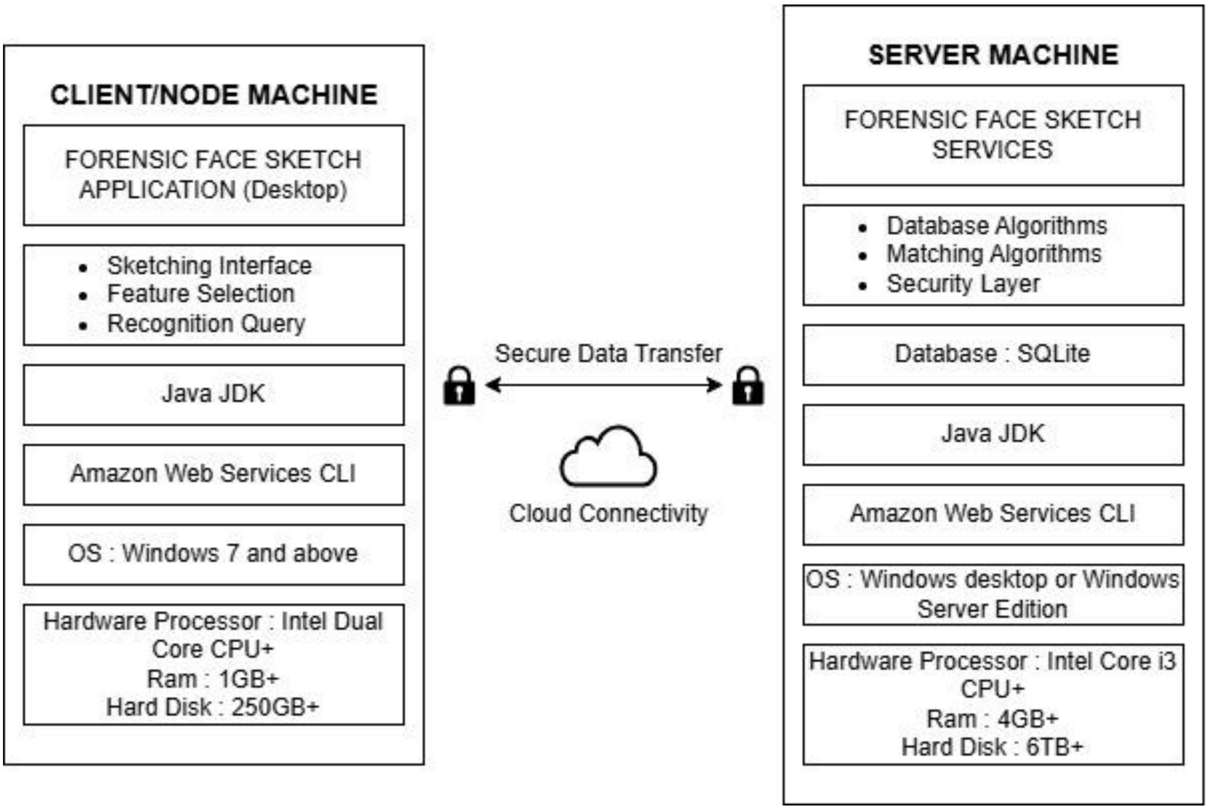


FIGURE 4.1.1 Architecture Diagram of the Application

4.1.2 Proposed Methods

Our application would be majorly used by the Law Enforcement Departments in order to reduce the overall time required to bring the criminal to justice and even to enhance the workforce and speed up the system by keeping accuracy in mind. So, keeping this scenario in mind the platform is designed to be as simple as possible in order to make sure that a user can create a sketch in the application without a formal training.

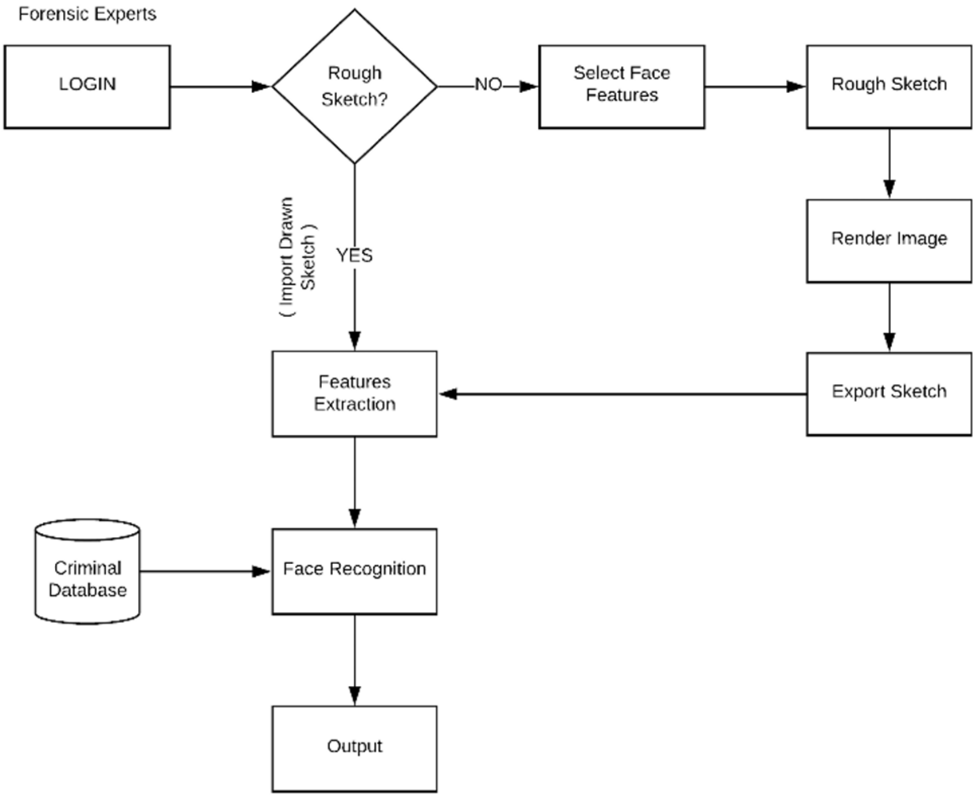


FIGURE 4.1.2 System Flow Chart of the Application

The above flowchart represents the overall flow of the system starting with the login page to the actual results been displayed after the sketch is been matched by the records in the database. The privacy and security are been kept in mind from the very first stage itself starting with the login page itself, the login page consists of two parts. At the start the login page fetches the Mac Address along with IP Address and HDD ID which is then been matched with the data been collected while installing the platform in the host machine and if the data does not match the platform would lock itself and won't allow the user to move further and use any feature of the platform.

Moving further, the second part consists of authenticating the user, which ensures that the user accessing the platform has complete privacy and security of their data and credentials. For this, the system verifies the user's login credentials entered on the platform and checks their authenticity before granting access. This mechanism ensures that only authorized users can access the system, maintaining data security and preventing unauthorized usage.

After the secure login on to the platform and moving further the platform uses something called as Backward Compatibility, this feature is been introduced in order to make a smooth transition from the current technique on to the new platform. The current technique been the use of hand drawn sketch been drawn by an expert forensic artist with years of experience and then the sketch been used by the law enforcement department to be showed on to various platforms in order to create a sense of awareness in people in order to find someone to recognize the suspect. So backward compatibility allows the law enforcement department to upload those hand drawn sketches on to the platform in order to use our face recognition module and match the suspect sketch with the large record and reducing the overall time and the efforts used in the previous age-old technique. If the law enforcement department doesn't have a hand drawn sketch and the law enforce department would wish to use the platform for creating a face sketch using our platform, they can access the canvas where they would find a wide range of facial elements in the database. The elements can be easily selected to create a described face sketch of the suspect and use the feature like drag and drop in order to arrange the elements according to the eye witness's description

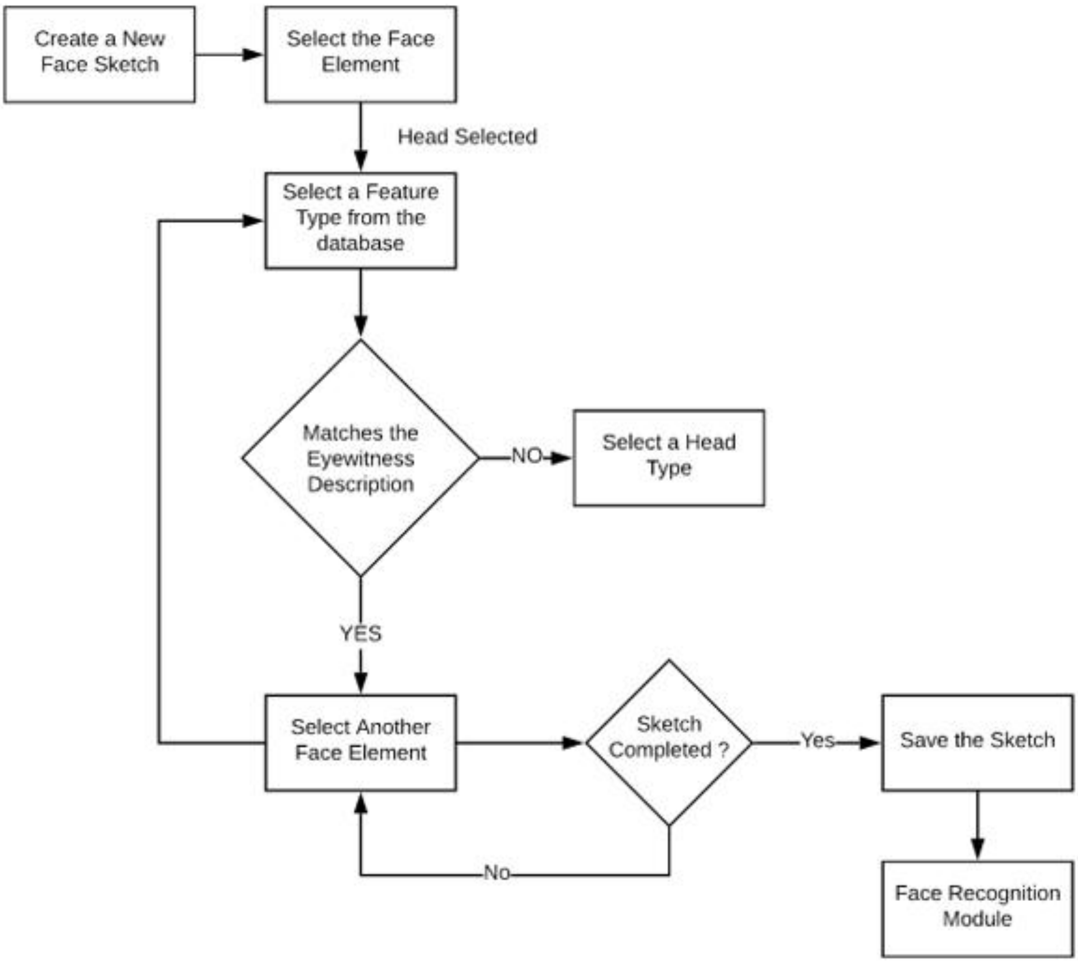


FIGURE 4.1.3. Flow Chart for Creating a sketch in the application

The platform is designed in such a way that one can use the platform without a prior professional training and knowledge of sketching. The user thus can select the main face category he/she wishes to select and would then prompt with a variety of option under that particular face category and then can select one feature based on the description provided by the suspect.

4.2.UML Diagrams

1. **Use Case Diagram**

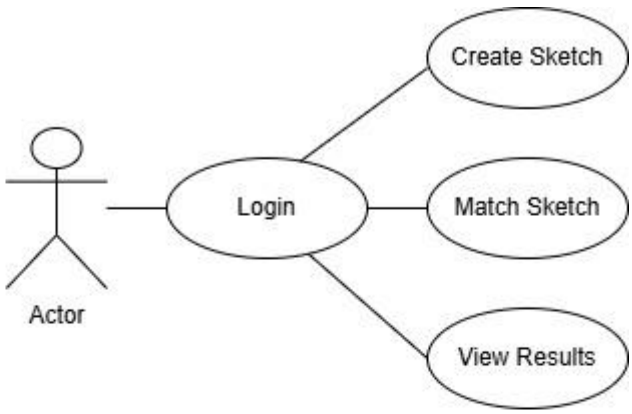


FIG 4.2.1 UseCase Diagram for facial sketch construction and recognition

- The actor (on the left) represents the end user, likely a law enforcement official or forensic expert.
- The Login use case is the starting point, indicating that the user must log in to access the system.
- After logging in, the user can perform three main actions:
    - **Create Sketch**: Build a new forensic face sketch using the platform’s tools.
    - **Match Sketch**: Use the created or uploaded sketch to search for matches within the criminal database.

- **View Results**: See the results of the sketch matching, including possible matches and details.

2. **Class Diagram**

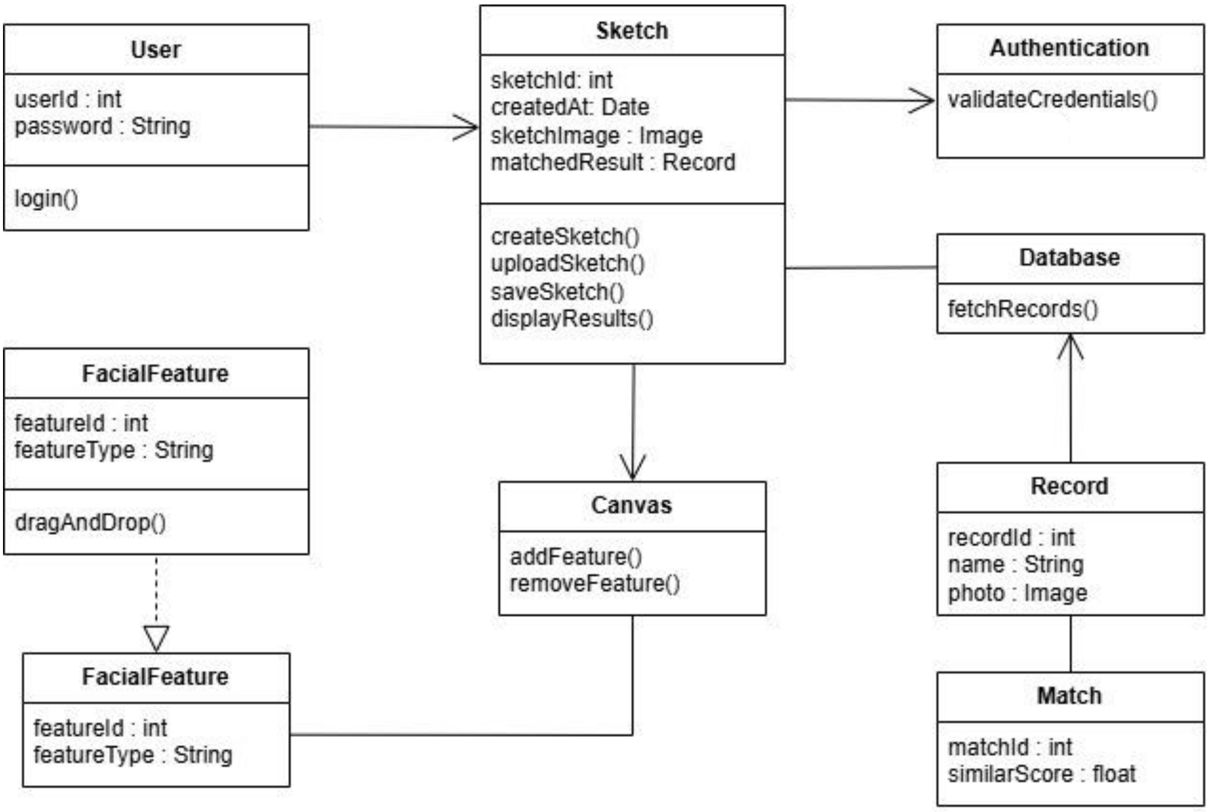


FIGURE 4.2.2.Class Diagram for facial sketch construction and recognition

It illustrates the main classes (or components), their attributes, methods, and the relationships between them:

- **User**: Contains userId and password, with a login() function to access the system.
- **Authentication**: Handles user verification by validating credentials and machine locking.
- **Sketch**: Stores details of the sketch (ID, creation date, image, matched results) and is responsible for creating, uploading, saving, and displaying the sketch through its functions (createSketch(), uploadSketch(), saveSketch(), displayResults()).

- **FacialFeature**: Represents facial components (with featureId and featureType) and supports drag-and-drop to add or modify features for the sketch on the canvas.

- **Canvas**: Works with the Sketch and FacialFeature classes to allow the addition and removal of features via addFeature() and removeFeature() methods, supporting interactive sketch construction.

- **Database**: Responsible for fetching records needed for matching, using the fetchRecords() method.

- **Record**: Holds criminal database entries (recordId, name, and photo) that are used for comparison with the sketch.

- **Match**: Represents the result of a matching operation, including matchId and similarScore.

3. **Sequence Diagram**

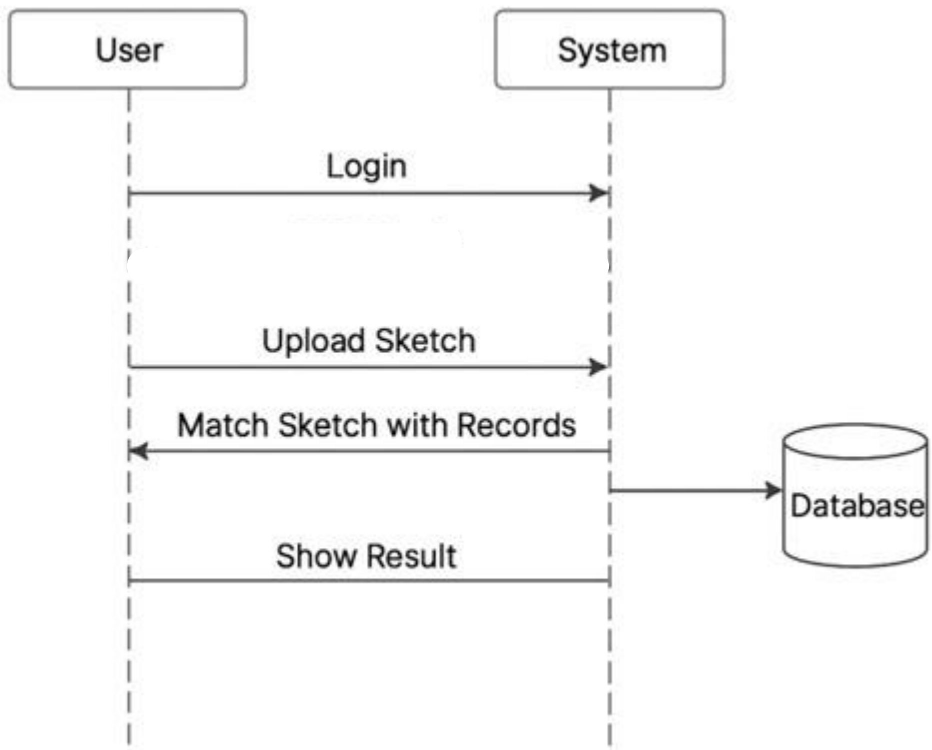


FIGURE 4.2.3. Sequence Diagram for facial sketch construction and recognition

- The user starts by sending a **Login** request to the system.
- Once the user is authenticated, they **Upload Sketch** to the system.
- The system then **matches the uploaded sketch with records** in the database.
- After processing, the system **shows the result** of the matching operation to the user.

4. **Deployment Diagram**

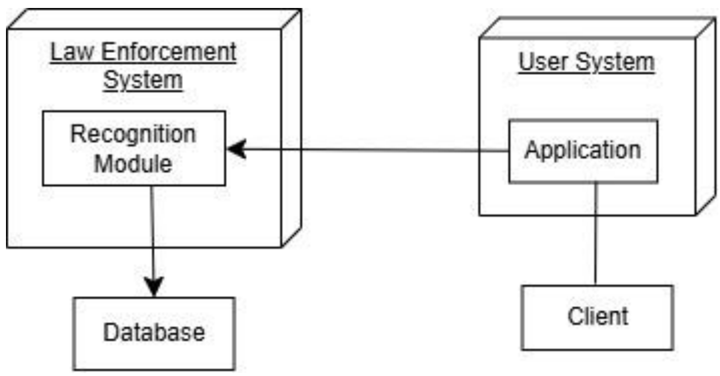


FIGURE 4.2.4. Deployment Diagram for facial sketch construction and recognition

- **User System**: Represents the end user's environment, consisting of the client device running the application. This is where users interact with the system—for creating, uploading, and viewing sketches.

- **Law Enforcement System**: Represents secure server infrastructure managed by law enforcement authorities. It hosts the Recognition Module, which processes uploaded sketches to identify matches against records in the connected database.

- **Database**: The backend repository storing all relevant data—including criminal records and images—that the Recognition Module queries for matching.

Connections:

- The User System (application) communicates securely with the Recognition Module in the Law Enforcement System.

- The Recognition Module directly accesses the Database for matching operations and result fetching.

4.3 Datasets and Technology Stack

Datasets

The proposed Face Sketch Recognition System requires a large set of paired sketch and photo images for model training and testing. The datasets used are publicly available benchmark datasets widely adopted in face sketch research:

1. **CUFS Dataset (Chinese University of Hong Kong Face Sketch Database)**
    - Contains 606 face photo–sketch pairs.
    - Includes images from the CUHK student set, AR, and XM2VTS databases.
    - Each photo has a corresponding hand-drawn sketch by professional artists.
    - Used for testing recognition accuracy and sketch synthesis algorithms.
2. **CUFSF Dataset (CUHK Face Sketch FERET Database)**
    - Extension of the CUFS dataset with 1,194 photo–sketch pairs.
    - Includes variations in lighting and expressions for realistic testing.
    - Suitable for training deep learning models that handle illumination and pose changes.
3. **Self-Collected Dataset (Optional)**
    - Contains manually created sketches and photos collected for testing under Indian demographic conditions.
    - Used to enhance regional accuracy and system validation.

Technology Stack

The Face Sketch Recognition System integrates advanced deep learning and secure computing methods. The following technologies and tools are used in development and deployment:

- **Frontend:** JavaFX for building a graphical user interface for sketch upload and recognition.
- **Backend:** Java for implementing core logic and sketch matching algorithms.
- **Deep Learning Framework:** TensorFlow/Keras for training CNN-based face recognition models.
- **Database:** MySQL or AWS RDS for storing facial features and metadata.
- **Cloud Platform:** AWS for centralized data storage and model deployment.
- **Security:** Machine Locking to ensure authorized access.
- **Networking:** Cloud-based centralized communication for real-time comparison.
- **Operating System:** Windows 10/11 for development and execution.

CHAPTER - 5

IMPLEMENTATION

5.1 Introduction to Implementation

This chapter presents the implementation details of the proposed Forensic Face Sketch Construction and Recognition System. It explains how the system is practically developed by integrating various modules, technologies, and functionalities described in the previous chapters. The implementation focuses on transforming the conceptual design into a working application that can be used by law enforcement agencies for efficient suspect identification. The system is implemented as a desktop-based application with centralized server support, ensuring both performance and security.

The core implementation is divided into multiple functional modules, including user authentication, face sketch construction, face recognition, database management, and cloud integration. Each module is developed to perform a specific task while maintaining seamless communication with other modules. The authentication module ensures secure access using machine locking. The face sketch construction module provides a user-friendly interface that allows users to create sketches using drag-and-drop facial features. The recognition module utilizes deep learning algorithms to extract features from the sketch and compare them with stored criminal records to identify possible matches. The system also integrates cloud services to handle computationally intensive tasks and maintain centralized data storage.

Additionally, this chapter includes descriptions of system interfaces, classes, and methods used in the implementation. Important code snippets are provided to illustrate key functionalities without exposing the complete source code. Screenshots of different system stages, such as login, sketch creation, and recognition results, are included to demonstrate the working of the application.

Overall, this chapter highlights how the proposed system is implemented in a structured and modular manner, ensuring scalability, security, and ease of use. The implementation bridges the gap between theoretical design and real-world application, making the system practical for deployment in forensic and law enforcement environments.

5.2 System Modules

5.2.1 Authentication Module

The Authentication Module is a critical component of the system that ensures secure access to the application. Since the system is designed for law enforcement usage, maintaining data security and preventing unauthorized access is of utmost importance. This module integrates three major security mechanisms: Login System and Machine Locking.

1. Login System:

The login system is the first step in accessing the application. Each authorized user is provided with a unique username and password. When the user enters the login credentials, the system validates them against the stored data in the database. If the credentials are correct, the user is allowed to proceed to the next level of authentication. If invalid credentials are entered, an error message is displayed and access is denied. This ensures that only registered users can attempt to access the system.

2. Machine Locking:

Machine locking is an additional security feature implemented to restrict the application to a specific system. The application captures unique hardware identifiers such as the Hard Disk ID (HDD ID) and MAC Address of the system during installation. These identifiers are stored securely and are checked every time the user attempts to log in.

If the system detects any mismatch in these hardware identifiers, it assumes that the application is being accessed from an unauthorized device and immediately blocks access. This prevents duplication, unauthorized installation, and misuse of the application on other systems.

The overall authentication process follows a multi-layered security approach:

Step 1: User enters username and password.

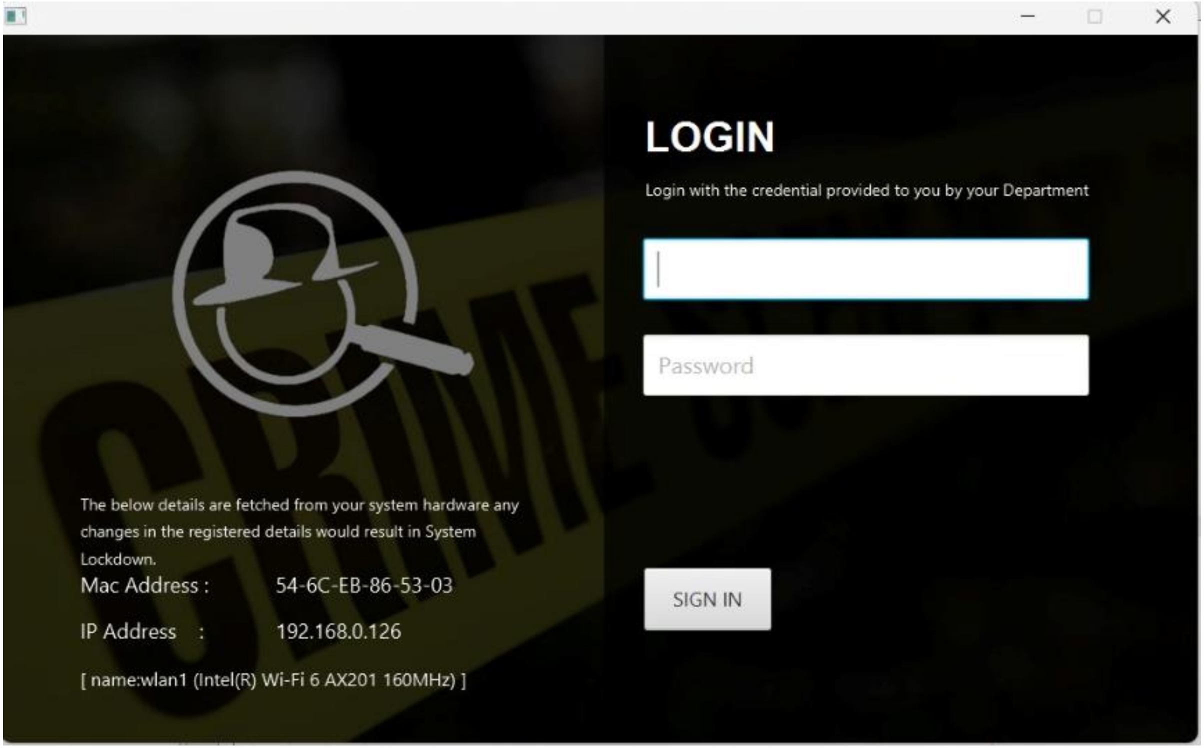
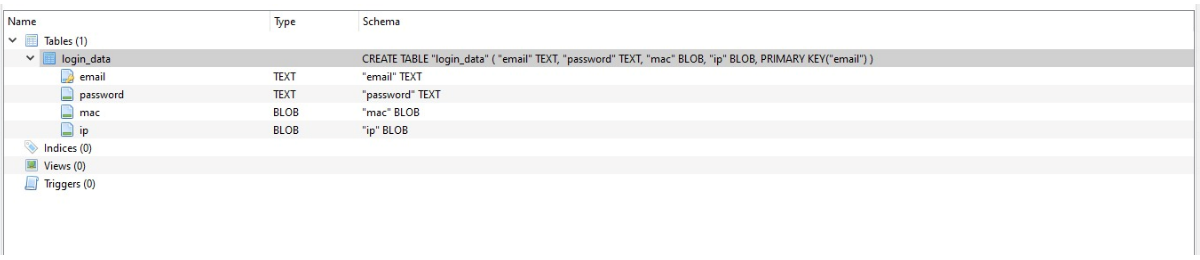


FIGURE 5.2.1(a). Login Page

Step 2: System validates credentials with database.



Name	Type	Schema
Tables (1)		
login_data		CREATE TABLE "login_data" ("email" TEXT, "password" TEXT, "mac" BLOB, "ip" BLOB, PRIMARY KEY("email"))
email	TEXT	"email" TEXT
password	TEXT	"password" TEXT
mac	BLOB	"mac" BLOB
ip	BLOB	"ip" BLOB
Indices (0)		
Views (0)		
Triggers (0)		

FIGURE 5.2.1(b). Login Data in Database

Step 3: Machine locking check is performed using HDD ID and MAC Address.

Step 4: If all validations are successful, access to the application dashboard is granted.

FIGURE 5.2.1(c). Dashboard

If any step fails, the system denies access and displays an appropriate error message.

This multi-level authentication mechanism ensures high security, protecting sensitive data and restricting system usage only to authorized personnel and approved devices.

5.2.2 Face Sketch Construction Module

The Face Sketch Construction Module is one of the core components of the system. This module enables users to create a composite face sketch of a suspect based on the description provided by an eyewitness. The main objective of this module is to provide a simple, user-friendly interface that allows even non-professional users to construct accurate face sketches without requiring artistic skills. The module is designed using a graphical user interface where different facial components are organized systematically. It mainly consists of a canvas area, feature selection panel, drag-and-drop functionality, and sketch saving options.

1. Canvas:

The canvas is the central working area of the application where the face sketch is constructed. It acts as a drawing board where all selected facial features are placed and arranged. When a user selects a particular facial feature, it is displayed on the canvas. The canvas allows multiple features to be combined together to form a complete face sketch. All component such as eyes, nose, lips, and head are layered properly on the canvas to maintain correct facial structure.

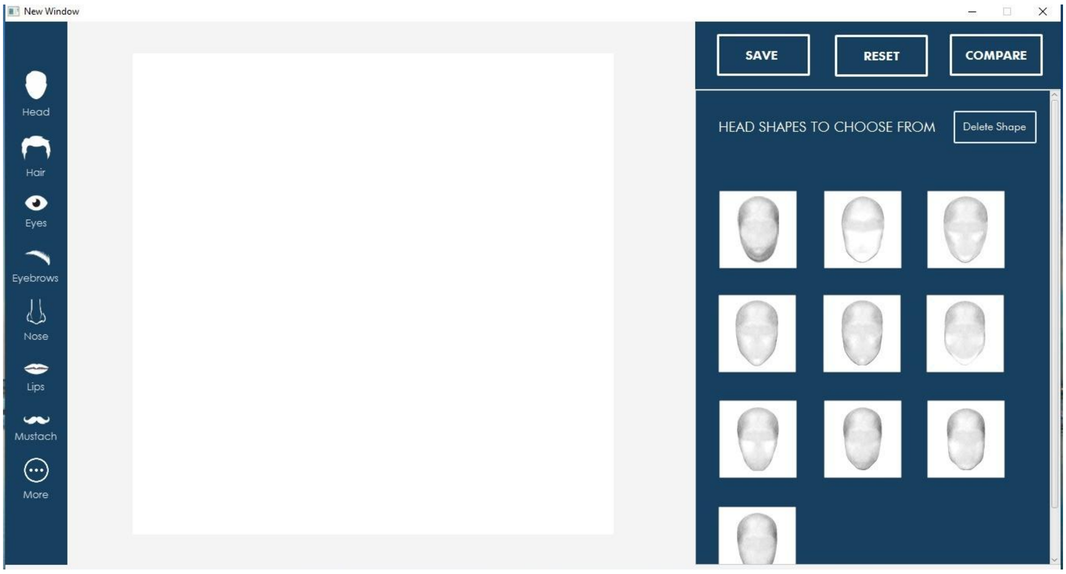


FIGURE 5.2.2(a). Canvas Page

2. Feature Selection:

The system provides a predefined library of facial features categorized into different groups such as head, eyes, nose, lips, ears, and hair. These features are displayed in a side panel for easy access. When a user selects a category, multiple variations of that feature are shown. The user can choose the most appropriate feature based on the eyewitness description. This organized categorization helps in faster and more accurate sketch creation.

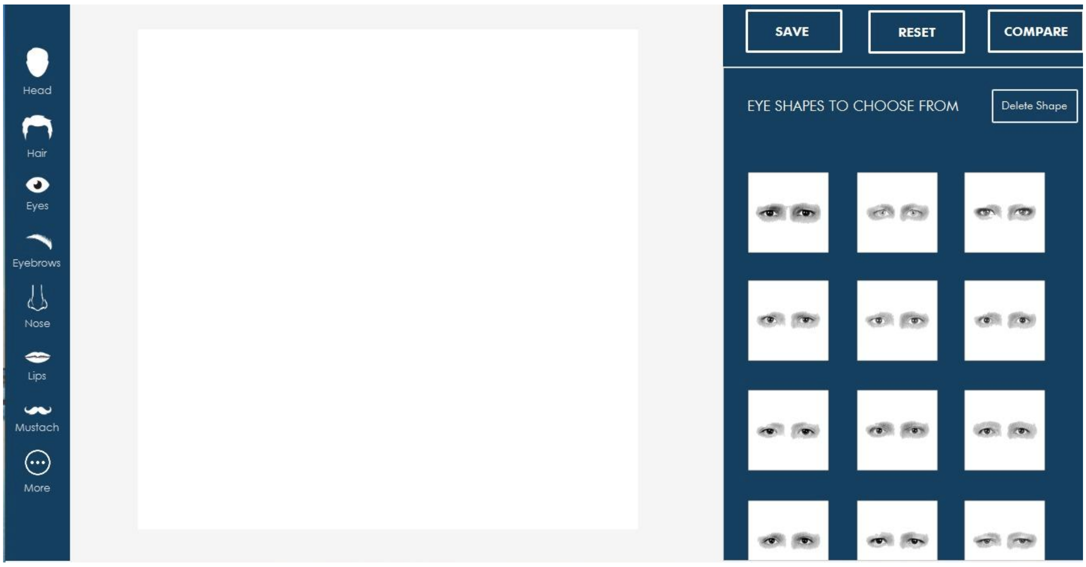


FIGURE 5.2.2(b). Features to sketch

3. Drag and Drop Functionality:

The selected facial features can be adjusted using drag-and-drop functionality. Users can move the features across the canvas to position them correctly. Additionally, features can be resized and repositioned to match the exact proportions of the suspect's face. This functionality improves flexibility and allows users to refine the sketch for better accuracy.

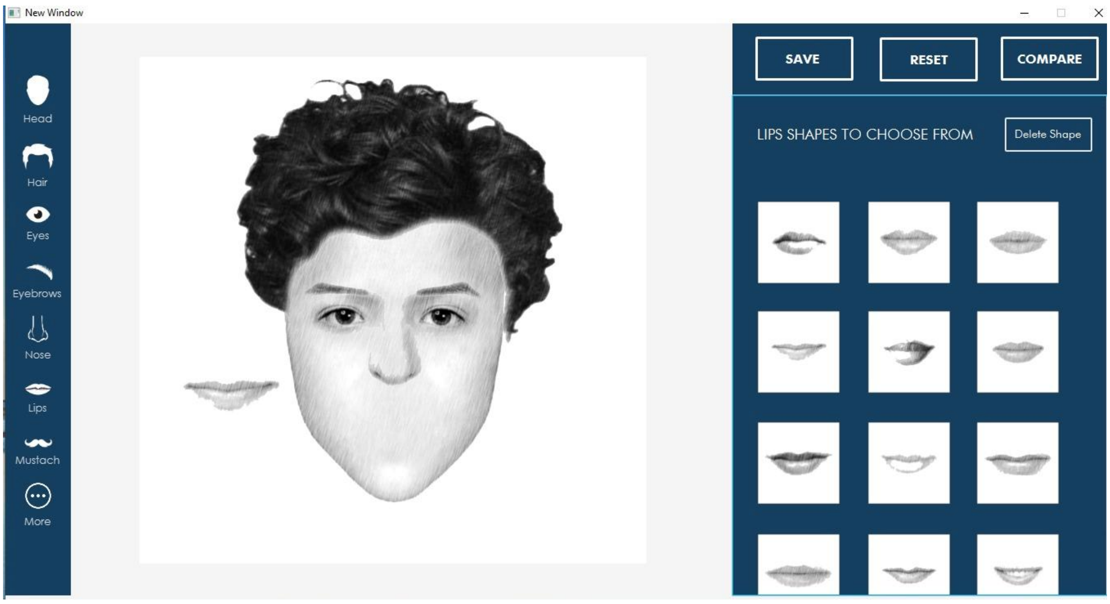


FIGURE 5.2.2(c). Features Drag & Drop Functionality

4. Editing Options:

The module provides editing features that allow users to modify the sketch easily. Users can replace any selected feature if it does not match the description. There is also an option to remove individual elements or clear the entire canvas. These options help in correcting mistakes and improving the overall quality of the sketch.

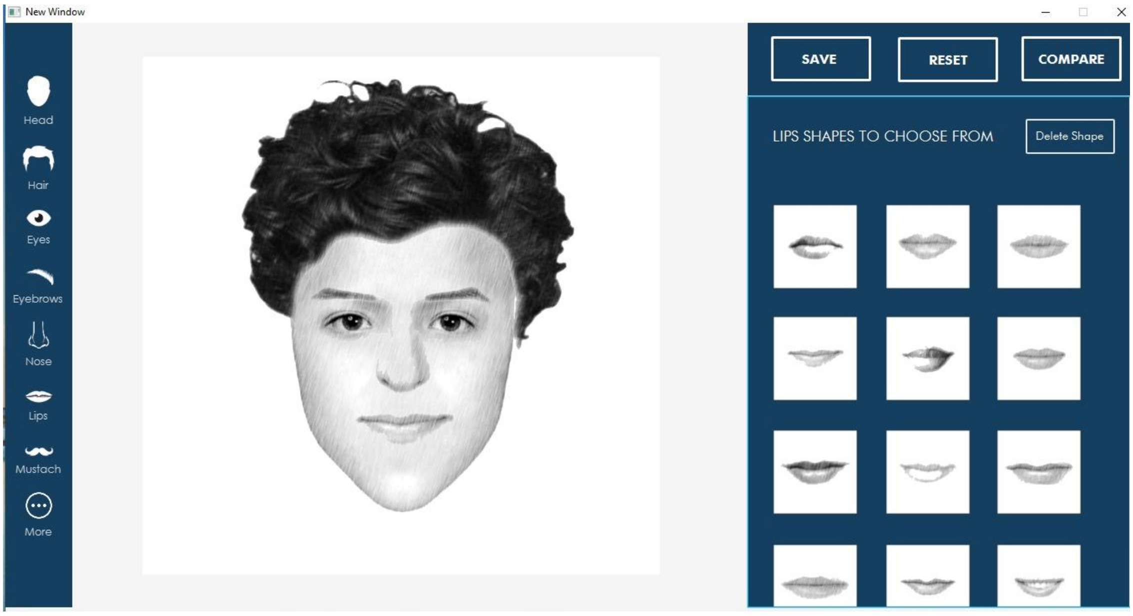


FIGURE 5.2.2(d). Editing Features

5. Save Option:

Once the sketch is completed, the user can save it in standard image formats such as PNG or JPG. The saved sketch can be stored either locally on the system or on the centralized server. This saved image can later be used for recognition or shared with other departments for investigation purposes.

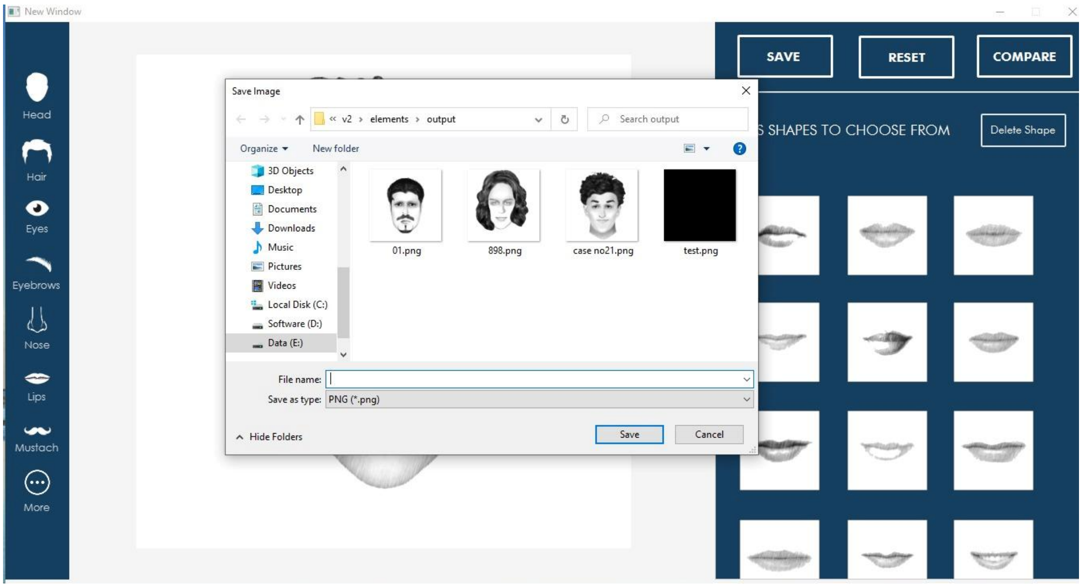


FIGURE 5.2.2(e). Saving sketch

6. Working Flow of Sketch Construction:

The process of constructing a face sketch follows a step-by-step approach:

Step 1: User selects a facial category (e.g., eyes, nose, head).

Step 2: System displays multiple feature options under the selected category.

Step 3: User selects a suitable feature based on eyewitness description.

Step 4: Selected feature is placed on the canvas.

Step 5: User adjusts the position and size using drag-and-drop.

Step 6: Steps are repeated for all facial components.

Step 7: Final sketch is reviewed and edited if necessary.

Step 8: Completed sketch is saved for further processing.

This structured approach ensures that the user can efficiently construct an accurate and realistic face sketch.

Overall, the Face Sketch Construction Module provides an intuitive and efficient way to create suspect sketches. It reduces dependency on forensic artists and enables faster sketch creation, which is crucial in time-sensitive criminal investigations.

The use of categorized feature selection and drag-and-drop interaction significantly improves usability and reduces the time required for sketch creation.

5.2.3 Face Recognition Module

The Face Recognition Module is responsible for identifying the suspect by matching the constructed or uploaded face sketch with the existing criminal database. This module uses deep learning techniques to analyse and compare facial features efficiently and accurately.

1. Upload Sketch:

The recognition process begins by uploading the face sketch into the system. The sketch can either be created using the application or uploaded as a hand-drawn image. The system accepts multiple image formats such as JPG and PNG. Once uploaded, the image is prepared for further processing.

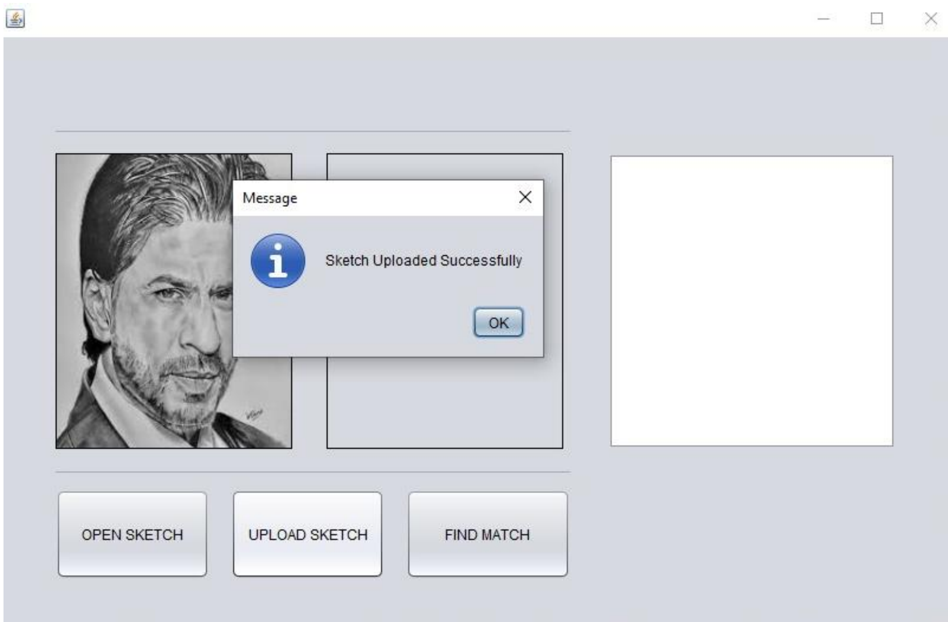


FIGURE 5.2.3(a). Upload Sketch

2. Feature Extraction:

After uploading the sketch, the system extracts important facial features from the image. This involves identifying key facial components such as eyes, nose, lips, and facial structure. The image is divided into smaller regions, and features are mapped using deep learning algorithms. This step converts the image into a feature vector that can be compared with stored data.

3. Matching Algorithm:

The extracted features are compared with the features of face images stored in the criminal database. The system uses deep learning models to calculate similarity between the input sketch and database images. The algorithm evaluates multiple records and identifies the closest match based on feature similarity.

4. Output Display:

After comparison, the system displays the matched results on the screen. The output includes:

- Matched face image

- Similarity percentage

- Confidence level

- Additional suspect details (if available)

The system presents results in a structured format for easy understanding by the user.

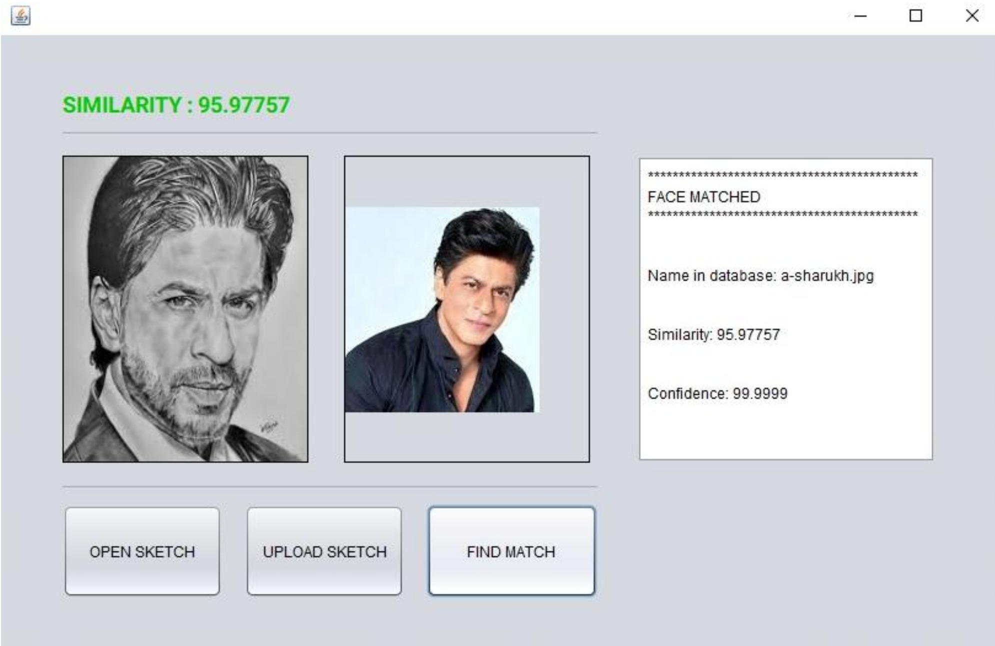


FIGURE 5.2.3(b). Output for Sketch Recognition

5. Similarity Percentage Explanation:

The similarity percentage indicates how closely the input sketch matches with a face in the database. It is calculated based on the similarity between extracted feature vectors. A higher percentage indicates a stronger match. For example, a similarity score above 85% is considered a strong match, while lower values indicate weaker similarity.

6. Working Flow:

Step 1: User uploads or selects a face sketch.

Step 2: System preprocesses the image.

Step 3: Facial features are extracted using deep learning.

Step 4: Features are compared with database records.

Step 5: Matching results are generated.

Step 6: Output is displayed with similarity percentage and details.

This module plays a vital role in identifying suspects quickly and accurately, significantly reducing investigation time.

5.3 Interfaces, Classes and Methods

This section describes the key interfaces, classes, and methods used in the implementation of the system. The application is developed using an object-oriented approach, where each component is designed as a separate class or interface to ensure modularity, maintainability, and scalability.

1. Interfaces:

a) User Interface (JavaFX UI):

The User Interface is developed using JavaFX. It provides an interactive graphical environment where users can perform operations such as login, sketch creation, and recognition. The UI is designed to be simple and user-friendly, allowing even non-technical users to interact with the system easily.

b) Database Interface:

The Database Interface handles communication between the application and the database. It provides methods to store, retrieve, and update user data and criminal records. This interface ensures smooth data flow between modules and maintains data consistency.

2. Classes:

a) UserLogin Class:

This class manages user authentication. It validates login credentials entered by the user and checks them against the stored database records. It also controls access to the system based on successful authentication.

b) SketchBuilder Class:

This class handles the face sketch construction process. It manages the canvas, feature selection, and placement of facial components. It allows users to add, remove, and adjust features to build a complete face sketch.

c) RecognitionEngine Class:

This class performs the face recognition process. It extracts features from the input sketch and compares them with stored images in the database. It calculates similarity scores and returns the best match.

d) DatabaseManager Class:

This class manages database operations such as storing user data, retrieving criminal records, and managing image storage. It ensures secure and efficient data handling.

3. Methods:

a) loginUser():

This method is used to validate user credentials. It checks the entered username and password against the database. If valid, it allows the user to proceed to uploading page.

b) uploadSketch():

This method allows users to upload a face sketch (either drawn or digital). It processes the image and prepares it for feature extraction.

c) extractFeatures():

This method extracts important facial features from the uploaded sketch using image processing and deep learning techniques.

d) matchFace():

This method compares extracted features with database records. It calculates similarity scores and identifies the closest match.

e) saveSketch():

This method saves the constructed face sketch in image format (PNG/JPG) either locally or on the server.

These interfaces, classes, and methods collectively ensure smooth functioning of the system by dividing the application into manageable and reusable components.

5.4 Code Snippets

This section presents selected code snippets that demonstrate key functionalities of the system. These snippets provide an overview of how important operations are implemented.

1. User Login Validation:

```
// LOGIN FUNCTION
private String Login() {

    try {
        conn = DriverManager.getConnection("jdbc:sqlite:login.sqlite");
    } catch (SQLException e) {
        System.out.println(e.getMessage());
    }

    String status = "Success";
    String e_mail = email.getText();
    String pass = password.getText();

    if(e_mail.isEmpty() || pass.isEmpty()) {
        error.setText("Empty credentials");
        status = "Error";
    }
    else {

        String sql = "SELECT * FROM login_data WHERE email = ? AND password = ?";

        try {

            preparedStatement = conn.prepareStatement(sql);
            preparedStatement.setString(1, e_mail);
            preparedStatement.setString(2, pass);

            resultSet = preparedStatement.executeQuery();

            if (!resultSet.next()) {
                error.setText("Enter Correct Email / Password");
                status = "Error";
            }
            else {
                loginerror.setText("successful");
            }

        } catch (SQLException ex) {
            System.err.println(ex.getMessage());
            status = "Exception";
        }
    }

    return status;
}
```

This method checks whether the entered username and password are valid.

2. Upload Sketch:

```
//UPLOAD THE SKETCH TO S3 BUCKET TO FIND A MATCH FROM AWS COLLECTION
private void upload_sketchActionPerformed(java.awt.event.ActionEvent evt) {
    Regions clientRegion = Regions.AP_SOUTH_1;
    String bucketName = "thirdeyepicsproj123";
    //String stringObjKeyName = "stringkey";
    String fileObjKeyName = "test.jpg"; //File name to be shown in S3 Bucket
    String fileName = sketch_path.getText();

    try {
        //This code expects that you have AWS credentials set up per:
        // https://docs.aws.amazon.com/sdk-for-java/v1/developer-guide/setup-credentials.html
        AmazonS3 s3Client = AmazonS3ClientBuilder.standard()
                .withRegion(clientRegion)
                .build();


        // Upload a file as a new object with ContentType and title specified.
        PutObjectRequest request = new PutObjectRequest(bucketName, fileObjKeyName, new File(fileName));
        ObjectMetadata metadata = new ObjectMetadata();
        metadata.setContentType("plain/text");
        metadata.addUserMetadata("title", "someTitle");
        request.setMetadata(metadata);
        s3Client.putObject(request);

        JOptionPane.showMessageDialog(null, "Sketch Uploaded Successfully");

        match_path.setText(null); //Clears the old path data of the image
        match.setIcon(null); //Clears the Old Matched Image
        match_similarity.setText(null); //Clears the Similarity Percentage
        match_properties.setText(null);
```

This method handles the uploading of sketch images into the system.

3. Feature Matching (Basic Logic):

```
//FIND MATCH FROM THE AWS COLLECTION AND SHOW RESULTS
private void find_matchActionPerformed(java.awt.event.ActionEvent evt) {
    String collectionId = "Records";
    String bucket = "thirdeyepicsproj123";
    String photo = "test.jpg";

    AmazonRekognition rekognitionClient = AmazonRekognitionClientBuilder.defaultClient();

    ObjectMapper objectMapper = new ObjectMapper();

    // Get an image object from S3 bucket.
    com.amazonaws.services.rekognition.model.Image image=new com.amazonaws.services.rekognition.model.Image()
            .withS3Object(new S3Object()
                    .withBucket(bucket)
                    .withName(photo));

    // Search collection for faces similar to the largest face in the image.
    SearchFacesByImageRequest searchFacesByImageRequest = new SearchFacesByImageRequest()
            .withCollectionId(collectionId)
            .withImage(image)
            .withFaceMatchThreshold(70F)
            .withMaxFaces(2);

    SearchFacesByImageResult searchFacesByImageResult =
            rekognitionClient.searchFacesByImage(searchFacesByImageRequest);


    List < FaceMatch > faceImageMatches = searchFacesByImageResult.getFaceMatches();
```

This method compares features of the input sketch with database features to calculate similarity.

4. Save Sketch:

```
@FXML //Save the sketch to image
private void onSave(MouseEvent event) {
    save_img(); //save image
}

@FXML //Open the upload and compare page
private void onCompare(MouseEvent event) {
    save_img(); //Save image

    // open the upload the sketch page
        try {
            FXMLLoader fxmlLoader = new FXMLLoader();
            fxmlLoader.setLocation(getClass().getResource("upload_sketch.fxml"));
            Scene scene = new Scene(fxmlLoader.load());
            Stage stage = new Stage();
            stage.setTitle("New Window");
            stage.setScene(scene);
            stage.resizableProperty().setValue(false); //Disable maximize button
            stage.show();
            ((Node)(event.getSource())).getScene().getWindow().hide();

        } catch (IOException e) {
            Logger logger = Logger.getLogger(getClass().getName());
        }
}
```

This method saves the constructed sketch into the system.

These code snippets highlight the core implementation logic of authentication, sketch handling, and face recognition processes.

CHAPTER - 6

TESTING

6.1 Testing Overview

The system was tested by executing different modules with various input data to ensure correct functionality. Each feature of the application was verified using valid and invalid inputs. The testing was focused on checking how the system behaves under different conditions and whether it produces the expected output.

The major modules tested include login authentication,sketch construction, and face recognition. The system was also tested for handling incorrect inputs and edge cases.

6.2 Test Cases

S.No.	Module	Input	Expected Output	Actual Output	Result
1	Login	Valid username and password	Login successful	Login successful	Pass
2	Login	Invalid username or password	Error message displayed	Error message displayed	Pass
4	Sketch Construction	Select facial features	Features displayed on canvas	Features displayed on canvas	Pass
5	Sketch Construction	Drag and drop features	Features repositioned correctly	Features repositioned correctly	Pass
6	Save Sketch	Save button clicked	Sketch saved successfully	Sketch saved successfully	Pass
7	Face Recognition	Upload sketch image	Image processed	Image processed	Pass

8	Face Recognition	Valid sketch	Matching result with similarity %	Matching result displayed	Pass
9	Face Recognition	Random sketch	No match found	No match found	Pass

6.3 Testing Screenshots

The following screenshots show the execution of different test cases and validate the correct working of the system:

• Login success screen

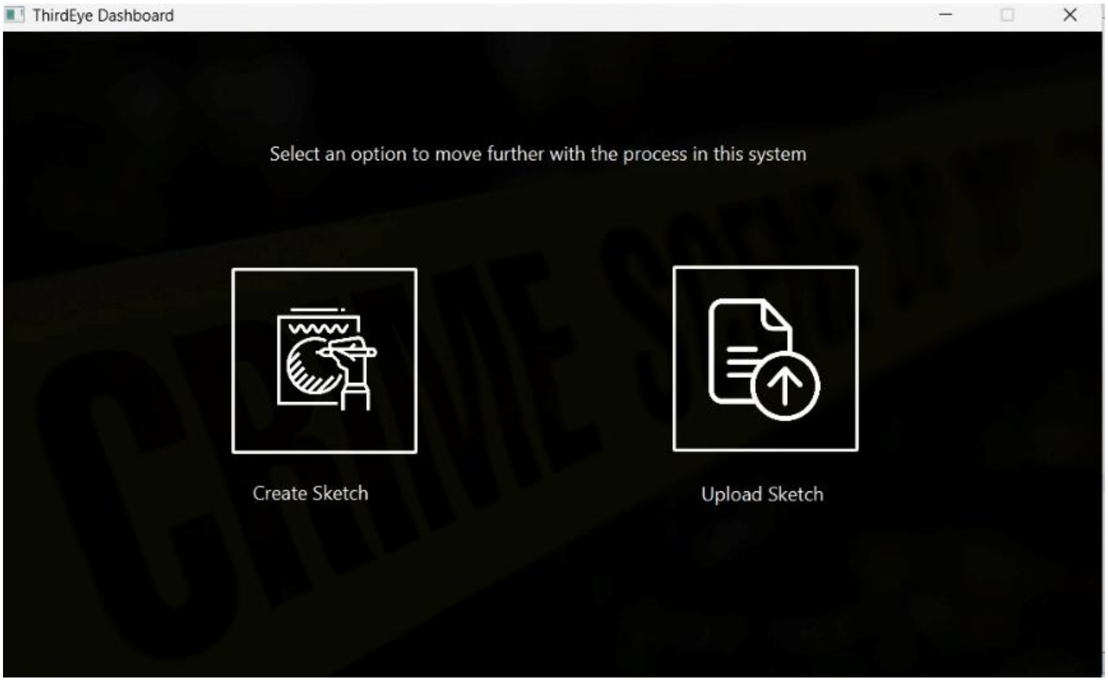


FIGURE 6.3.1. Login Success Page

• Login error message

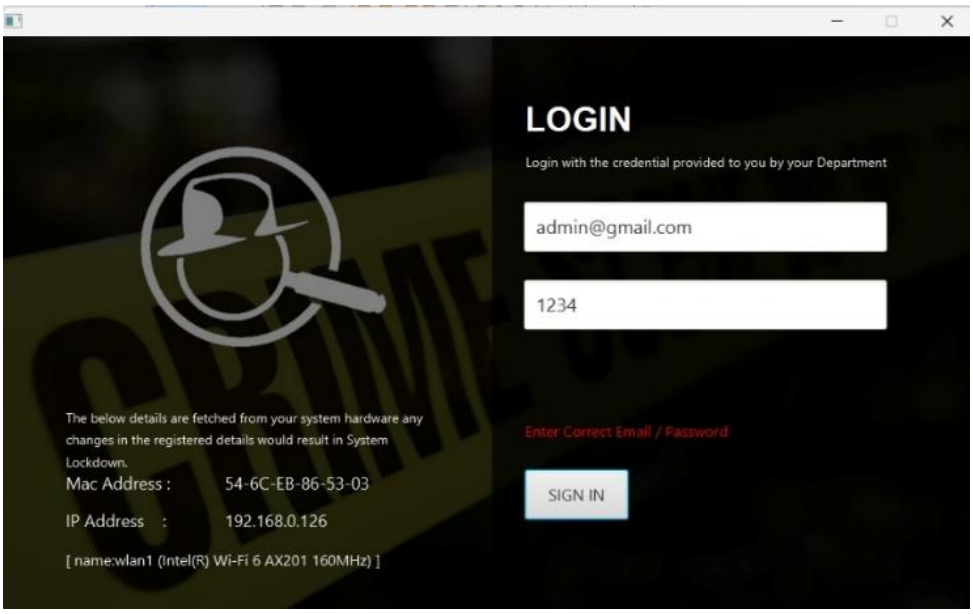


FIGURE 6.3.2. Login Error Page

• Face recognition result screen

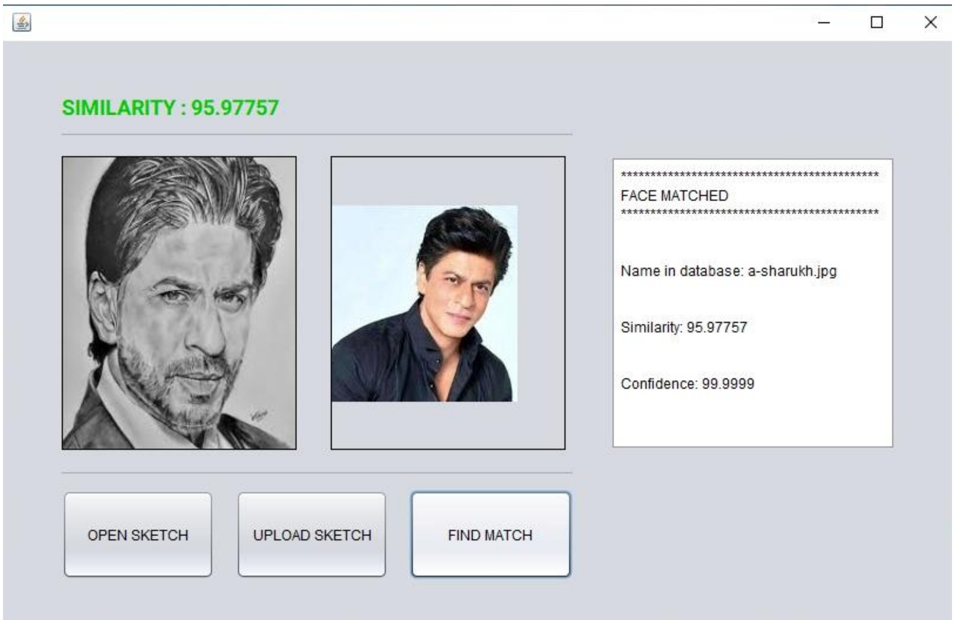


FIGURE 6.2.3(a). Sketch Recognition Testing

These screenshots confirm that the system is functioning correctly for all tested scenarios.

6.4 Testing Summary

The testing phase of the Forensic Face Sketch Construction and Recognition System was carried out successfully by executing all modules with different input scenarios. Each functionality of the system was thoroughly verified using both valid and invalid inputs to ensure correct behaviour under all conditions.

The login module was tested with multiple combinations of usernames and passwords to verify authentication accuracy. The system correctly restricted access when incorrect credentials were entered, confirming the effectiveness of the security mechanisms.

The face sketch construction module was tested by selecting various facial features, adjusting their positions using drag-and-drop functionality, and saving the final sketch. The module performed efficiently, allowing users to create sketches without any errors or delays.

The face recognition module was tested by uploading different types of sketches, including valid sketches and random images. The system successfully processed the input images, extracted features, and matched them with database records. It displayed accurate results along with similarity percentages, confirming the reliability of the recognition algorithm.

The system also handled edge cases such as invalid inputs and unmatched sketches effectively by displaying appropriate messages without crashing or producing incorrect results.

Overall, all the test cases were executed successfully, and the system produced the expected outputs in all scenarios. The testing results confirm that the application is stable, secure, accurate, and suitable for real-world deployment in law enforcement environments.

CONCLUSION

The Forensic Face Sketch Construction and Recognition System developed in this project successfully addresses the limitations of traditional forensic sketching methods by integrating modern technologies such as deep learning, cloud computing, and secure authentication mechanisms. The system provides a complete and practical solution for law enforcement agencies to construct and recognize suspect faces efficiently. The application enables users to create face sketches using an intuitive drag-and-drop interface without requiring professional artistic skills, significantly reducing dependency on forensic artists and enabling faster sketch creation during time-sensitive investigations. The constructed sketches, along with uploaded hand-drawn sketches, can be processed by the recognition module, which uses advanced feature extraction and matching algorithms to identify suspects from a criminal database.

One of the major strengths of the system is its ability to combine both traditional and modern approaches. It supports backward compatibility by allowing existing hand-drawn sketches to be uploaded and analysed, ensuring that previously collected data can still be effectively utilized in the digital system. The integration of security mechanisms ensures that only authorized users can access the application, which is crucial in law enforcement environments where data confidentiality and system integrity are critical. Additionally, the use of cloud infrastructure improves system scalability and performance by enabling faster processing of large datasets.

Testing results indicate that the system performs efficiently with high accuracy in face recognition, achieving reliable matching results with similarity percentages. The application also demonstrates stability, ease of use, and quick response times, making it suitable for real-world implementation.

In conclusion, the proposed system successfully bridges the gap between traditional forensic methods and modern AI-based technologies. It enhances the speed, accuracy, and security of criminal identification processes, making it a valuable tool for law enforcement agencies. The project demonstrates the practical application of artificial intelligence in solving real-world problems and lays a strong foundation for further advancements in forensic identification systems.

FUTURE ENHANCEMENT

The Project 'Forensic Face Sketch Construction and Recognition' is currently designed to work on very few scenarios like on face sketches and matching those sketches with the face photos in the law enforcement records. The platform can be much enhanced in the future to work with various technologies and scenarios enabling it to explore various media and surveillances medium and get a much wider spread and outputs, The platform can be modified to match the Face sketch with the human faces from the video feeds by using the 3D mapping and imaging techniques and same can be implemented to the CCTV surveillances to perform face recognition on the Live CCTV footage using the Face Sketch. The platform can further be connected to social media has social media platforms acts has a rich source for data in today's world, this technique of connecting this platform with the social media platform would enhance the ability of the platform to find a much more accurate match for the face sketch and making the process much more accurate and speeding up the process. In all the platform could have features which could be different and unique too and easy to upgrade, when compared to related studies on this field, enhancing the overall security and accuracy by standing out among all the related studies and proposed systems in this field.

ABBRIVIATIONS

CNN	Convolutional Neural Network
HMAC	Hashed Message Authentication Code
TOTP	Time-Based One Time Password
HOTP	HMAC-Based One Time Password
MAC Address	Media Access Control Address
HD ID	Hard Drive Identifier
IDE	Integrated Development Environment
CSS	Cascading Style Sheets
FXML	XML-based markup language for JavaFX
API	Application Programming Interface
ML	Machine Learning
SRS	Software Requirement Specification
IP	Internet Protocol
GPU	Graphics Processing Unit
CPU	Central Processing Unit
DFD	Data Flow Diagram
UML	Unified Modelling Language

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